ITA0610 - Machine Learning

1.Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Experiment: FIND-S Algorithm

Aim

To implement and demonstrate the **FIND-S algorithm** for finding the **most specific hypothesis** from a given set of training data samples.

Algorithm

1. Start with the most specific hypothesis **h**.
2. Read the training dataset.
3. For each training example:
    - If the example is **positive (Yes)**:
        - Compare each attribute with the hypothesis.
        - If the hypothesis attribute is empty, replace it with the training value.
        - If the values differ, replace that attribute with **'?'**.
    - Ignore negative examples.
4. Repeat until all training examples are processed.
5. Display the final hypothesis.

Python Program

```
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
```

```
]

hypothesis = ['0'] * (len(data[0]) - 1)

print("Initial Hypothesis:")
print(hypothesis)
for example in data:
    if example[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = '?'

print("\nFinal Hypothesis:")
print(hypothesis)
```

OUTPUT:

```
[1]
0s
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

hypothesis = ['0'] * (len(data[0]) - 1)

print("Initial Hypothesis:")
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for example in data:
    if example[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = '?'

print("\nFinal Hypothesis:")
print(hypothesis)

Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

The FIND-S algorithm was successfully implemented. It identified the most specific hypothesis that correctly classifies all positive training examples.